CFO AI ASSISTANT

Version 2.0

Your Bank's Financial Brain ■

Imagine having a super-smart assistant that knows EVERYTHING about your bank's money— from how much cash you have, to whether you're following all the rules, to what your competitors are doing. That's what we built! ■

- ■ **Instant answers to financial questions**
- ■ **5 specialized AI agents working 24/7**
- ■ **Real-time regulatory compliance monitoring**
- ■ **Predictive forecasting & risk alerts**

Designed & Developed By:
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The Problem ■

Picture this: You're a CFO at a Nigerian bank. Every day, you're drowning in questions...

Your CEO asks:

"What's our capital adequacy ratio RIGHT NOW?" You scramble through spreadsheets for 30 minutes.

The Board wants to know:

"Are we compliant with CBN regulations?" You need to check 15 different reports.

A Manager questions:

"Can I approve this ■25 million expense?" You dig through policy documents.

Your team inquires:

"What are competitors doing?" You spend hours researching.

Auditors demand:

"Show me NPL trends for the past 6 months." You export data from 3 systems.

■ Result: Hours wasted. Decisions delayed. Stress levels through the roof!

The Solution ■

CFO AI Assistant v2.0 is like having a genius financial expert sitting next to you, ready to answer ANY question about your bank in seconds—not hours!

Here's what makes it AWESOME:

■■ Natural Conversation

Just ask questions like you're texting a friend: "What's our liquidity ratio?" or "Can I approve this expense?"

■ Lightning Fast

Get answers in under 3 seconds. No more waiting for reports or searching through files.

■ Always Accurate

Pulls real data from your bank's database. No guessing, no mistakes.

■ Multiple AI Experts

5 specialized agents handle different tasks—SQL for data, RAG for policies, Compliance for regulations, Market Intelligence for competition, and Financial Analyst for trends.

■ Beautiful Dashboard

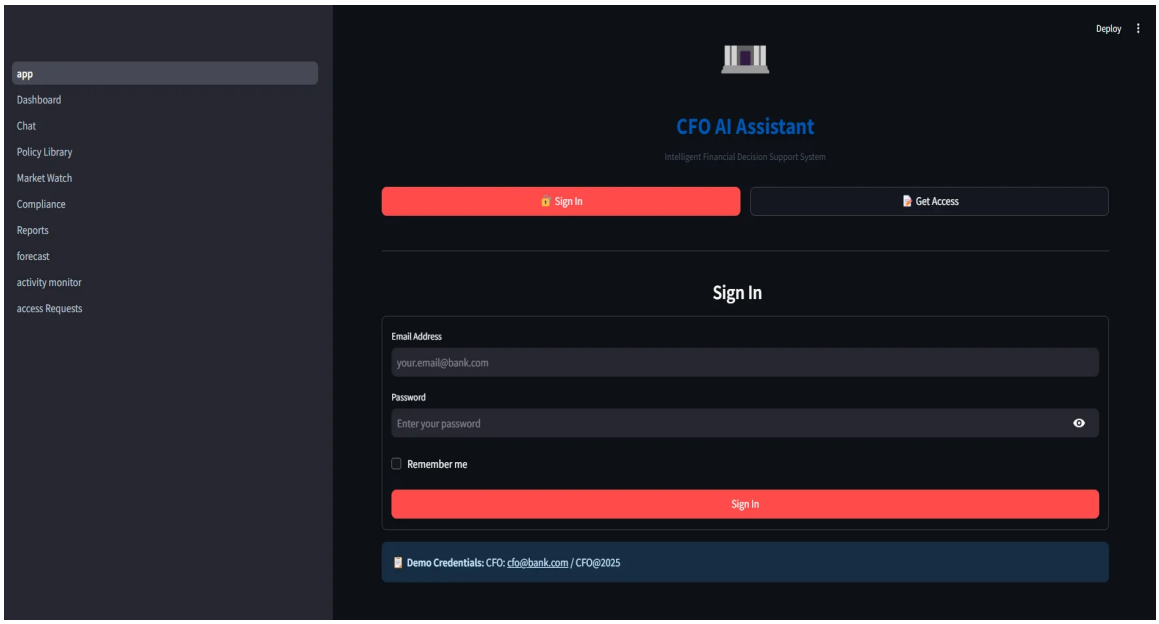
See all your key metrics at a glance—Capital Adequacy, Liquidity, Profitability, Asset Quality.

■ Bank-Grade Security

Role-based access control. CFOs see everything, managers see relevant data, analysts get read-only access.

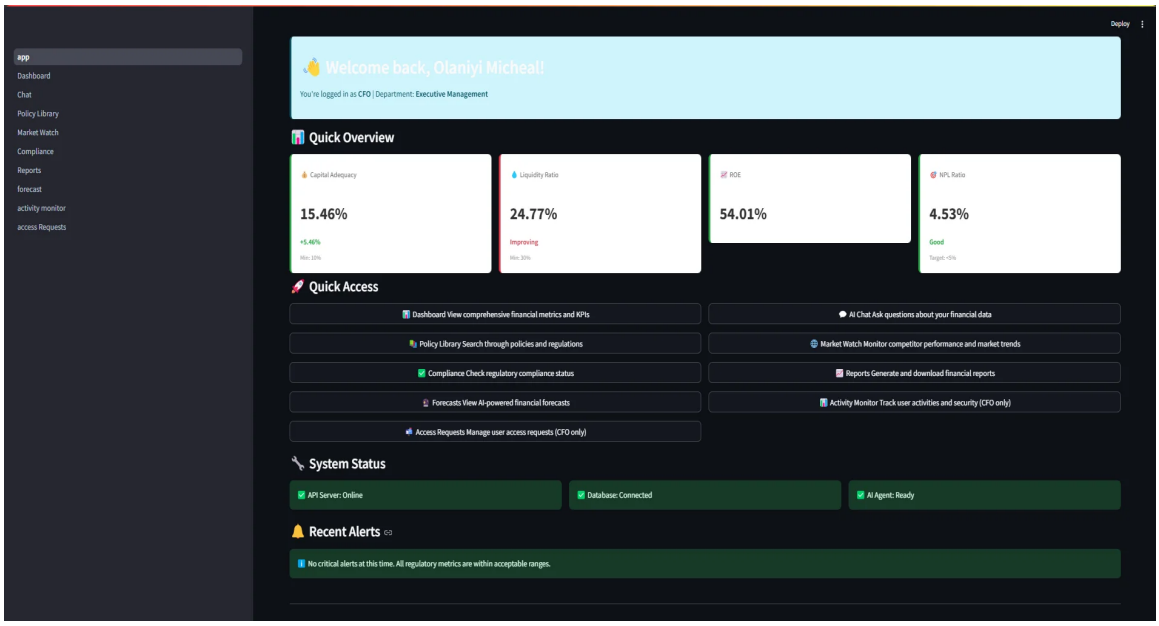
System Login & Access Control ■

Secure authentication with role-based access. Users can request access, and CFOs approve or deny with one click.



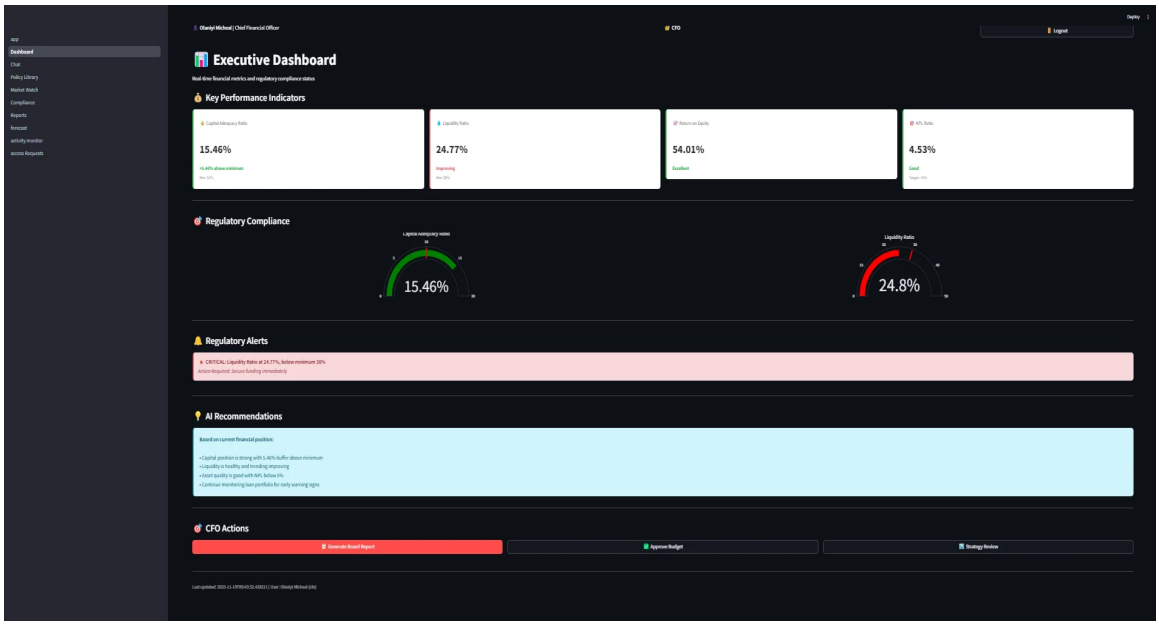
Executive Dashboard - Quick Overview ■

The homepage gives CFOs instant access to all key metrics and quick navigation to specialized modules. Capital Adequacy, Liquidity Ratio, ROE, and NPL Ratio are displayed prominently with status indicators.



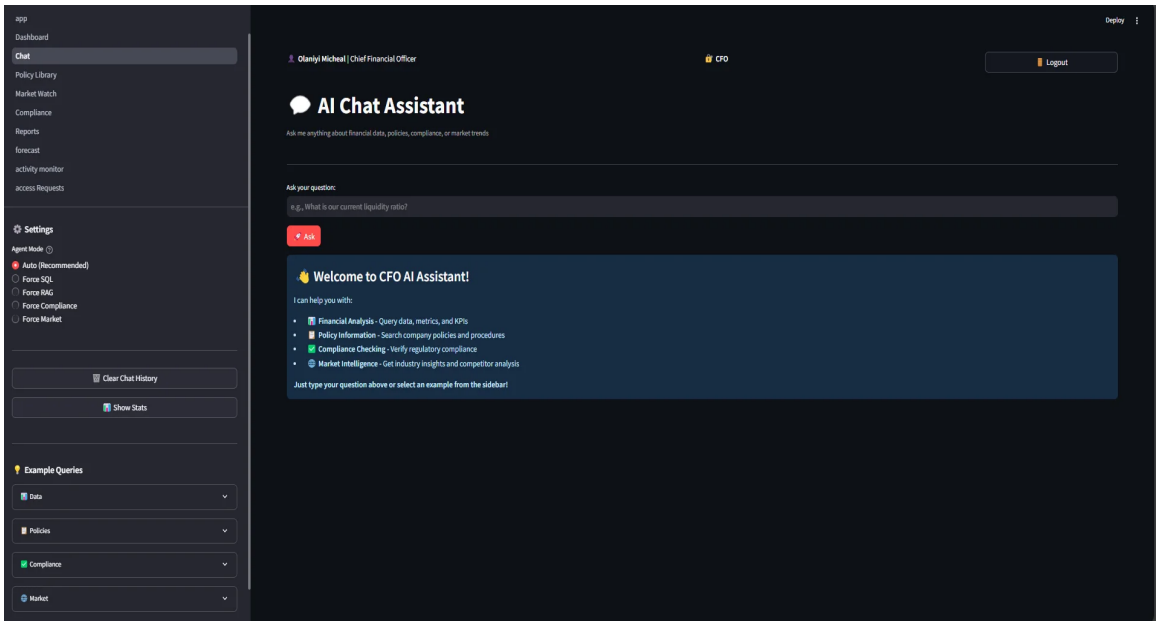
Complete Dashboard View ■

The full executive dashboard shows KPIs with visual gauges for regulatory compliance. Real-time alerts notify CFOs of any metrics approaching critical thresholds. AI-powered recommendations help with strategic decision-making.



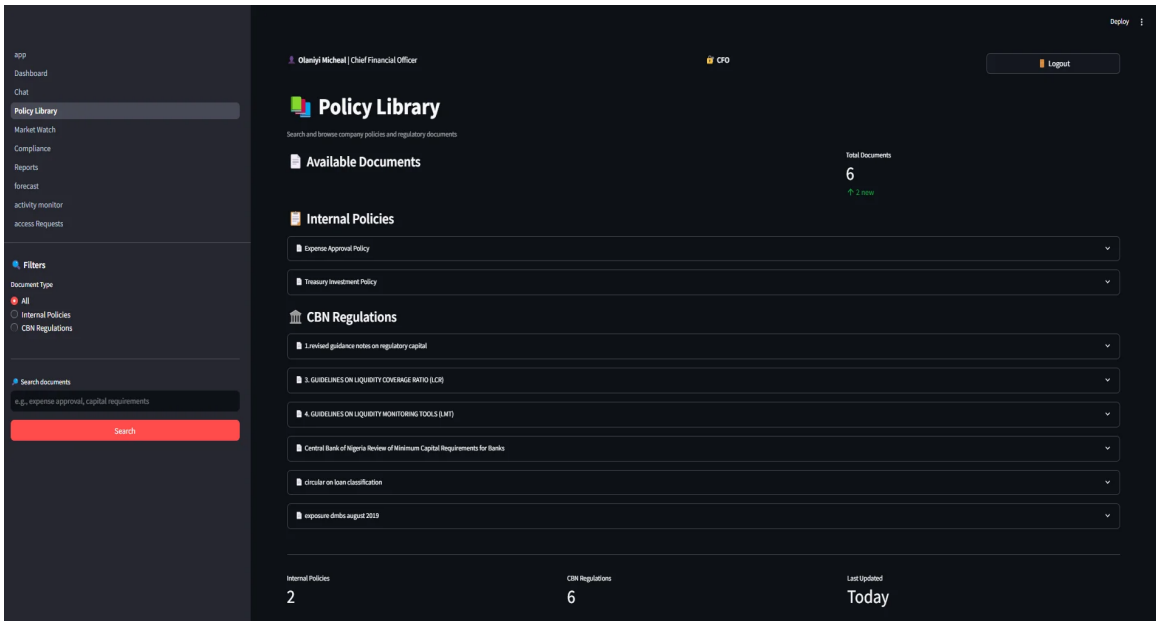
AI Chat Assistant ■

Ask questions in natural language. The AI understands context, remembers your conversation, and provides accurate answers with source citations. You can force specific agents or let the system automatically route your question to the right expert.



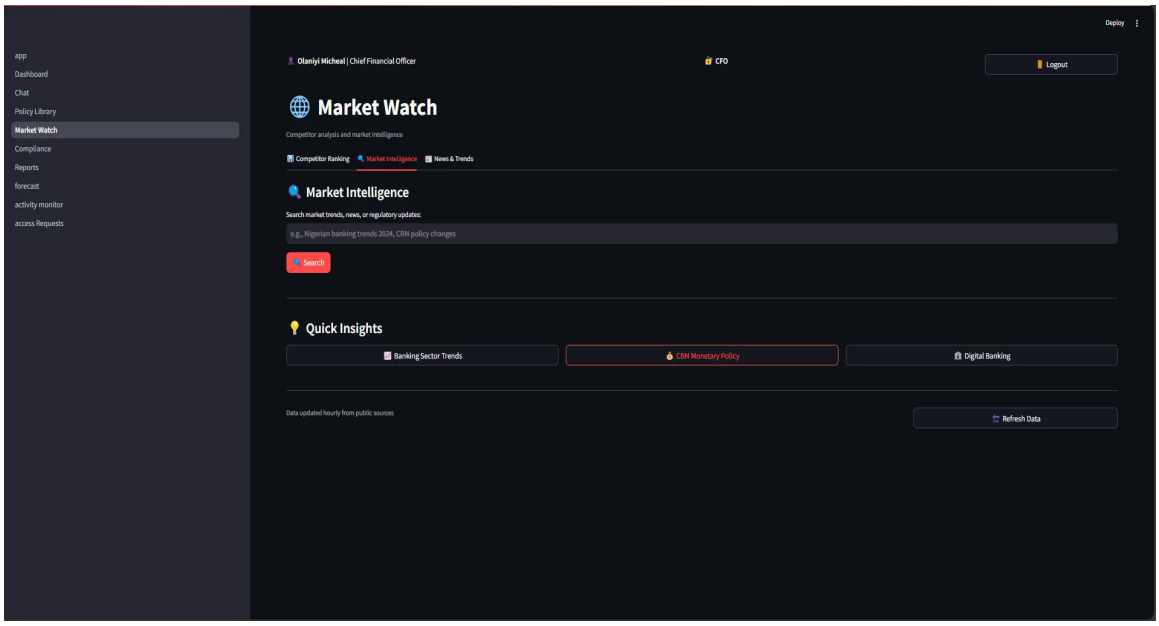
Policy Library ■

All your internal policies and CBN regulations in one searchable location. The AI can instantly retrieve and cite specific sections, making policy compliance easy. Documents are categorized and filterable for quick access.



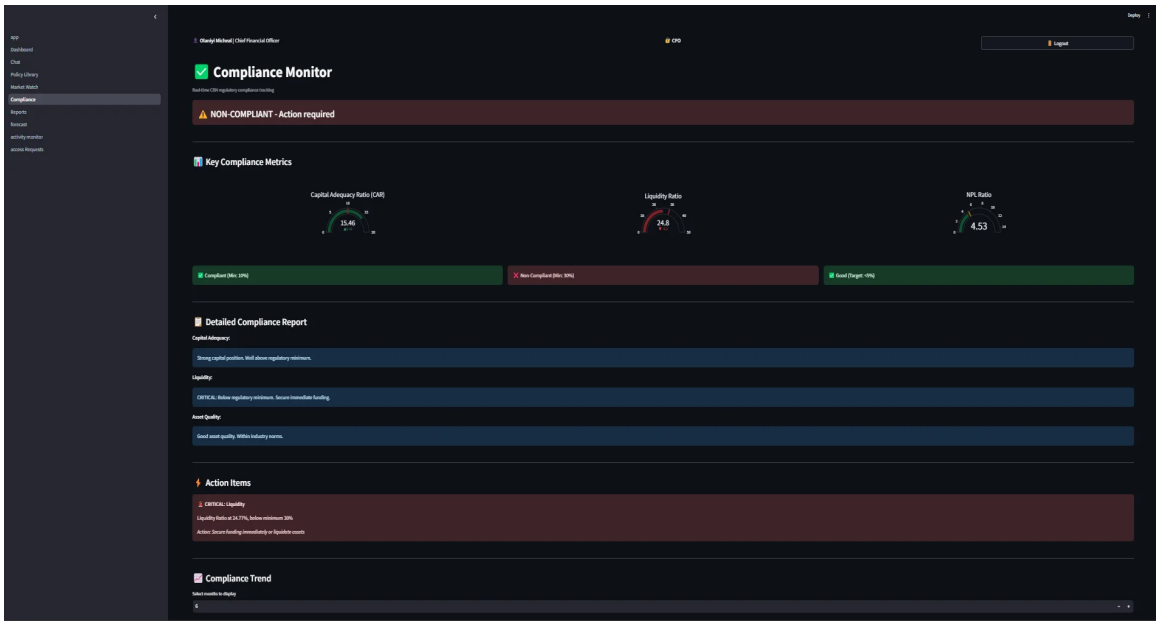
Market Watch ■

Monitor competitor performance and stay updated on market trends. The Market Intelligence agent searches the web for current information about other banks, interest rates, regulatory changes, and industry news.



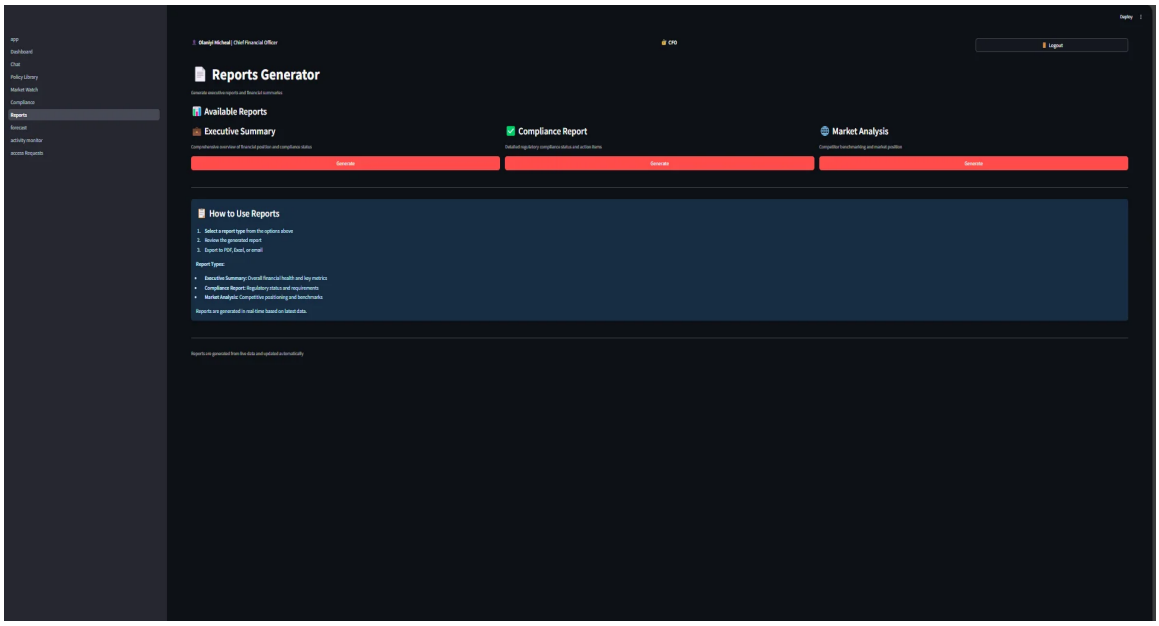
Compliance Monitor

Real-time monitoring of CBN regulatory requirements with visual gauges for key ratios. Detailed compliance reports show exactly where you stand, with specific action items for any violations. Historical trends help you track improvement over time.



Report Generator ■

Generate professional reports with one click. Choose from Executive Summary, Compliance Report, or Market Analysis. Reports are automatically formatted and can be exported as PDF or Excel files.



Financial Forecasting ■

AI-powered predictions for key financial metrics using statistical models and historical data. See forecasts for Profit After Tax, Return on Equity, and other critical indicators. Confidence intervals show the range of possible outcomes.



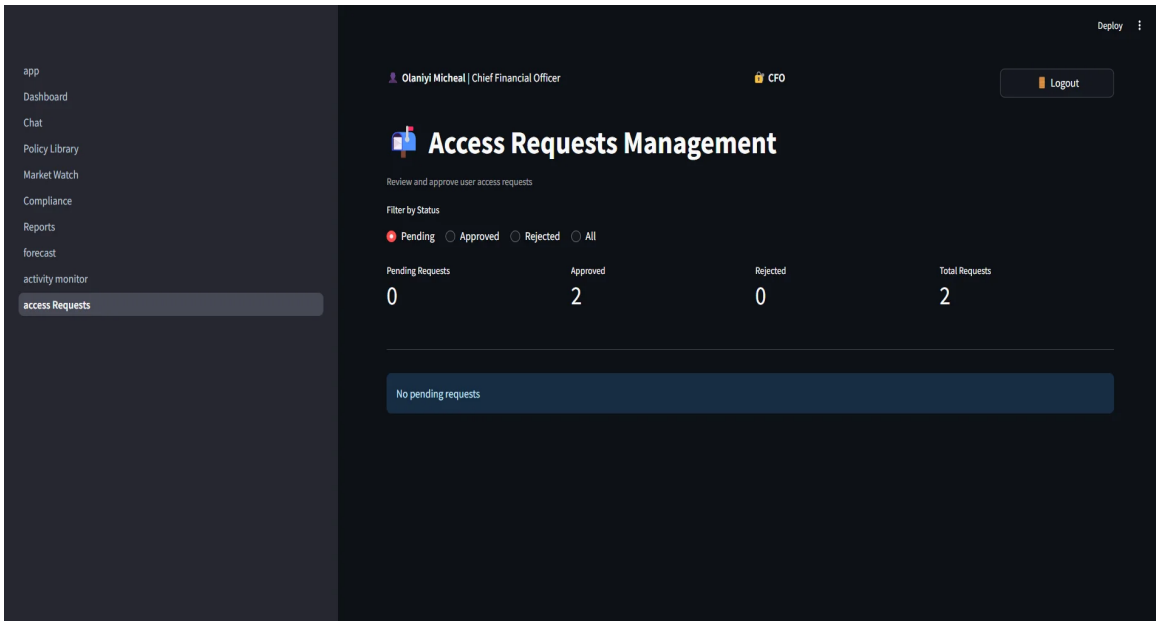
Activity Monitor ■

Track all user activities for security and audit purposes. See who's using the system, what actions they're taking, and when. Filter by date, user, or activity type. Export activity logs for compliance reporting.



Access Request Management ■

Employees can request system access directly through the interface. CFOs review requests with full context (name, employee ID, department, role requested, and justification). Approve or deny with one click, and the system automatically sends notifications.



Real-World Scenarios ■

Let's see how CFO AI Assistant handles real situations at your bank:

Scenario 1: Board Meeting in 10 Minutes ■

The Board wants latest financial figures, but your analyst is out sick.

Old way: Panic. Scramble through files. Show up with outdated numbers.

With CFO AI: Ask "Generate a board report with all key metrics." Get a complete PDF in 30 seconds. Walk in confident. ■

Scenario 2: CBN Audit Next Week ■

You need to prove compliance with all regulations.

Old way: Spend 2 days pulling reports from different systems, checking each ratio manually, hoping you didn't miss anything.

With CFO AI: Ask "Show me our complete compliance status." Get instant report showing all ratios, violations (if any), and recommendations. ■

Scenario 3: Manager Needs Expense Approval ■

IT Manager wants to buy ■35M worth of servers. Can you approve?

Old way: Search for expense policy document. Try to remember thresholds. Call Finance team for confirmation.

With CFO AI: Ask "Can I approve a ■35M IT expense?" Get instant answer with policy reference: "Requires VP and CFO approval." ■

Scenario 4: Strategy Planning ■

You need to know if you can expand lending without hurting capital ratios.

Old way: Build complex Excel models. Run multiple scenarios. Takes days and might still have errors.

With CFO AI: Ask "If we increase loans by ■5B, what happens to our CAR?" Get instant calculation with forecast and risk assessment. ■

Ready to Transform Your Bank? ■

CFO AI Assistant v2.0 is not just software—it's your competitive advantage. While other banks struggle with spreadsheets and manual processes, you'll have instant answers, perfect compliance, and data-driven decisions.

What You Get:

- ■ Complete system with all 5 AI agents
- ■ Beautiful web dashboard accessible from any device
- ■ Full setup and integration with your database
- ■ Training for your finance team
- ■ 6 months of support and updates
- ■ Customization to match your specific policies
- ■ Regular AI model updates as technology improves

■ Investment: Contact for custom pricing based on your bank's size
■ ROI: Recover costs within 3 months through time savings alone
■ Implementation: 2-4 weeks from contract to launch

Next Steps:

1■■ Schedule a 30-minute demo to see it in action

2■■ We'll do a free pilot with your financial data

3■■ Your team tests it for 2 weeks

4■■ If you love it (you will!), we deploy to production

Let's Connect! ■

I'm passionate about using AI to solve real-world problems in the financial sector. Whether you're a bank looking to modernize, or an organization seeking AI engineering expertise, I'd love to hear from you!

Olaniyi Micheal Oluwatosin
AI Engineer | Financial Technology Specialist

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■ WhatsApp: https://wa.me/2348174223657
■ Location: Lagos, Nigeria

■ Open to International Collaboration
■ Available for Remote AI Engineering Projects

"The best time to upgrade your bank was yesterday. The second best time is NOW."

Join the AI Revolution in Nigerian Banking ■■■

PART 2

Technical Deep Dive

For Developers, Data Scientists, and Technical Decision Makers

This section provides in-depth technical documentation for developers, data scientists, CTOs, and technical decision-makers who need to understand the architecture, implementation details, and integration requirements of the CFO AI Assistant system.

System Architecture ■■

CFO AI Assistant follows a microservices architecture pattern with clear separation of concerns between the frontend, backend API, AI agents, and data layer.

High-Level Architecture:

Frontend Layer: Streamlit-based web application with multiple pages for different functionalities

API Layer: FastAPI REST API handling all client requests with CORS support

Orchestration Layer: Query router and orchestrator coordinating between specialized agents

Agent Layer: 5 independent AI agents each with specific capabilities

Data Layer: PostgreSQL database with optimized schema for financial data

AI Model Layer: Claude Sonnet 4 by Anthropic (via API) for natural language processing

Request Flow:

1. User submits query via Streamlit frontend
2. Frontend sends HTTP POST to FastAPI backend (/api/query endpoint)
3. Orchestrator receives query and invokes Query Router
4. Router analyzes query intent and selects appropriate agent
5. Selected agent executes with access to database and AI model
6. Agent returns structured response with answer, metadata, and sources
7. Orchestrator packages response and sends back to frontend
8. Frontend displays formatted response to user

System Organogram ■


CFO AI ASSISTANT - SYSTEM ORGANOGRAM
■■■

Technology Stack ■

Core Technologies:

Programming Language: Python 3.10+

Why: Chosen for rich ecosystem of data science, AI, and web libraries. Type hints improve code quality.

AI Model: Claude Sonnet 4 (Anthropic API)

Why: 200,000 token context window allows processing entire policy documents. Superior reasoning for financial analysis.

Backend Framework: FastAPI 0.104+

Why: Async support, automatic API documentation, data validation with Pydantic, high performance.

Frontend Framework: Streamlit 1.28+

Why: Rapid development of data apps, built-in components for charts/tables, Python-native.

Database: PostgreSQL 14+

Why: ACID compliance, complex queries, JSON support, proven reliability for financial data.

Containerization: Docker & Docker Compose

Why: Consistent deployment across environments, easy scaling, isolated services.

Python Libraries & Dependencies:

anthropic==0.7.7: Official Anthropic SDK for Claude API integration

fastapi==0.104.1: Web framework for building APIs

uvicorn==0.24.0: ASGI server for running FastAPI

streamlit==1.28.1: Frontend framework for dashboard

psycopg2-binary==2.9.9: PostgreSQL adapter for Python

sqlalchemy==2.0.23: SQL toolkit and ORM

pydantic==2.5.0: Data validation using Python type annotations

python-jose==3.3.0: JWT token creation and validation

passlib==1.7.4: Password hashing library

pandas==2.1.3: Data manipulation and analysis

plotly==5.18.0: Interactive charting library

python-multipart==0.0.6: For file upload support

aiofiles==23.2.1: Async file operations

httpx==0.25.1: Async HTTP client for external API calls

Database Schema ■■

The database is designed for optimal performance in financial analytics with proper indexing and relationships. All tables include audit timestamps and are normalized to 3NF.

Core Tables:

balance_sheet

Purpose: Assets, liabilities, and equity. Updated quarterly.

Columns: id, report_date, total_assets, total_liabilities, shareholder_equity, cash_and_equivalents, loans_and_advances

profit_and_loss

Purpose: Income statement data. Updated monthly.

Columns: id, report_date, total_revenue, operating_expenses, profit_before_tax, profit_after_tax, net_interest_income

regulatory_ratios

Purpose: CBN compliance metrics. Calculated daily.

Columns: id, report_date, capital_adequacy_ratio, liquidity_ratio, npl_ratio, loan_to_deposit_ratio

loan_portfolio

Purpose: Individual loan records. Updated on disbursement/repayment.

Columns: id, loan_id, borrower_name, loan_amount, interest_rate, disbursement_date, maturity_date, sector, status

competitor_data

Purpose: Market intelligence on other banks. Updated weekly.

Columns: id, competitor_name, metric_name, metric_value, data_date, source

users

Purpose: System users and authentication. Created on signup.

Columns: id, email, password_hash, role, name, employee_id, department, status, created_at

activity_log

Purpose: Audit trail of all user actions. Inserted on every action.

Columns: id, user_email, action_type, details, timestamp

access_requests

Purpose: Pending access requests. Created when users request access.

Columns: id, full_name, email, employee_id, department, requested_role, justification, status, created_at

Indexing Strategy:

- Primary keys on all id columns (auto-increment)
- Index on report_date columns for time-based queries
- Composite index on (user_email, timestamp) in activity_log
- Index on status column in access_requests for filtering
- Unique index on email in users table

AI Agents - Technical Implementation ■

All agents inherit from BaseAgent class which provides common functionality: logging, error handling, response formatting, and execution tracking.

1. SQL Agent (SQLAgent)

File: `agents/sql_agent.py`

Converts natural language to SQL queries using Claude's code generation capabilities.

Key Features:

- Query safety validation (prevents DROP, DELETE, UPDATE)
- Schema injection into prompts for accurate SQL generation
- Result formatting with automatic pandas DataFrame conversion
- Error handling with user-friendly messages

Prompt Strategy: Few-shot learning with database schema and example queries

2. Policy RAG Agent (PolicyRAGAgent)

File: `agents/policy_rag_agent.py`

Retrieval-Augmented Generation for policy questions using document search and synthesis.

Key Features:

- Document chunking and embedding (for future vector search)
- Keyword-based retrieval from policy database
- Citation of source documents in responses
- Conversation memory for follow-up questions

Prompt Strategy: Inject relevant policy excerpts into context before asking Claude

3. Compliance Agent (ComplianceAgent)

File: `agents/compliance_agent.py`

Monitors CBN regulatory requirements and generates compliance reports.

Key Features:

- Automated ratio calculations from database
- Threshold comparison against CBN regulations
- Severity classification (critical, high, medium, low)
- Actionable recommendations for violations

Prompt Strategy: Provide Claude with current ratios and ask for professional compliance report

4. Market Intelligence Agent (MarketIntelligenceAgent)

File: `agents/market_intelligence.py`

Web search integration for competitor analysis and market trends.

Key Features:

- Integration with search APIs (planned)
- Current: Uses competitor_data table
- Structured output for comparison charts
- Source attribution for all facts

Prompt Strategy: Ask Claude to synthesize search results into actionable insights

5. Financial Analyst Agent (FinancialAnalystAgent)

File: `agents/financial_analyst.py`

Time-series analysis and forecasting using statistical models.

Key Features:

- Trend detection (moving averages, linear regression)

- Forecast generation (3-6 month horizons)
- Confidence intervals for predictions
- Anomaly detection in financial metrics

Prompt Strategy: Provide historical data and ask Claude to interpret trends and make predictions

Security & Deployment ■

Security Implementation:

Password Hashing: SHA-256 with salt. Passwords never stored in plain text.

Session Management: JWT tokens with expiry. Tokens stored in session state.

Role-Based Access Control: Middleware checks user role before page access. Three levels: CFO, Manager, Analyst.

SQL Injection Prevention: Parameterized queries. No string concatenation in SQL.

API Rate Limiting: FastAPI rate limiter prevents abuse (100 requests/minute per IP).

Audit Logging: All actions logged with timestamp, user, and details. Immutable append-only log.

CORS Configuration: Restricted to known origins. Credentials required for cross-origin requests.

Environment Variables: Secrets (API keys, DB passwords) stored in .env file, not in code.

Deployment Architecture:

The system uses Docker Compose for orchestration with three main services:

postgres: PostgreSQL 14 database. Data persisted in Docker volume. Port 5432 exposed internally.

backend: FastAPI application. Depends on postgres. Port 8000 exposed to host. Health checks every 30s.

frontend: Streamlit application. Depends on backend. Port 8501 exposed to host. Auto-restart enabled.

Deployment Steps:

1. Clone repository: `git clone [repo_url]`

2. Configure environment: Copy `.env.example` to `.env`, fill in API keys

3. Build containers: `docker-compose build`

4. Initialize database: `docker-compose run backend python scripts/init_db.py`

5. Start services: `docker-compose up -d`

6. Access frontend: `http://localhost:8501`

7. Access API docs: `http://localhost:8000/docs`

Production Considerations:

• Use reverse proxy (Nginx) for SSL termination and load balancing

• Configure database backups (daily full + hourly incremental)

• Set up monitoring (Prometheus + Grafana for metrics)

• Implement log aggregation (ELK stack or similar)

• Use managed PostgreSQL (AWS RDS, Azure Database, etc.) for HA

• Scale horizontally by adding more backend containers

• Cache frequently accessed data (Redis recommended)

• Regular security updates for all dependencies

API Documentation ■

Complete REST API with OpenAPI/Swagger documentation available at `http://localhost:8000/docs`

Core Endpoints:

GET `/`

API health check and available endpoints

GET `/api/health`

Service health status

POST `/api/query`

Main chat endpoint. Send user query, receive AI response.

GET `/api/dashboard/snapshot`

Current financial snapshot (all KPIs)

GET `/api/dashboard/alerts`

Regulatory alerts and warnings

GET `/api/compliance/status`

Current compliance status with all checks

GET `/api/compliance/trend`

Historical compliance data (6 months)

GET `/api/compliance/report`

Detailed compliance report (text)

GET `/api/forecasts`

Financial forecasts for key metrics

GET `/api/stats`

System usage statistics

POST `/api/reports/generate`

Generate PDF/Excel reports

GET `/api/auth/users`

List all users (admin only)

POST `/api/auth/create-user`

Create new user (admin only)

PUT `/api/auth/users/{email}`

Update user (admin only)

DELETE `/api/auth/users/{email}`

Deactivate user (admin only)

Example API Call:

```
import requests # Query the AI response = requests.post(
"http://localhost:8000/api/query", json={ "query": "What is our current
capital adequacy ratio?", "force_agent": None # Let router decide, or
specify "SQL", "RAG", etc. } ) result = response.json()
print(result['answer']) print(f"Agent used: {result['agent_used']}")
print(f"Sources: {result['sources']}")
```

Integration Guide ■

Integrating with Existing Bank Systems:

1. Database Connection

Update DATABASE_URL in `.env` to point to your existing PostgreSQL database. Or, set up ETL pipeline to sync data to CFO AI Assistant's database.

2. Schema Mapping

If your schema differs, update `core/database.py` with correct table and column names. Alternatively, create database views that match expected schema.

3. Authentication Integration

Replace built-in auth with your bank's SSO (SAML, OAuth). Implement custom authentication in `frontend/auth.py`.

4. Policy Documents

Upload your policies to `data/policies/` directory. Run `scripts/ingest_policies.py` to index them.

5. Customize Agents

Modify agent behavior by editing files in `agents/`. Add bank-specific logic, thresholds, or calculations.

6. Branding

Update logo, colors, and text in `frontend/components/`. Configure bank name in `core/config.py`.

Testing Strategy:

Unit Tests: Test individual agents and services. Run: `pytest tests/`

Integration Tests: Test API endpoints. Run: `pytest tests/integration/`

Load Testing: Use `locust` or `k6` to simulate concurrent users.

Security Testing: Run `bandit` for code security issues. Use `safety` to check dependencies.

User Acceptance Testing: Pilot with small group of users (5-10) before full rollout.

Maintenance & Support ■■

Regular Maintenance Tasks:

Daily: Monitor system logs for errors. Check API response times. Review failed queries.

Weekly: Update competitor data. Review and approve access requests. Check database size and performance.

Monthly: Update policy documents. Review and update financial thresholds. Analyze user feedback. Check for dependency updates.

Quarterly: Full system backup verification. Security audit. Performance optimization. User training sessions.

Monitoring Recommendations:

• API response time (target: <500ms for 95th percentile)

• Database query performance (slow query log)

• Agent execution time by type

• Error rate (target: <0.1%)

• User activity patterns (peak hours, most used features)

• Claude API usage and costs

• Database connection pool utilization

• Memory and CPU usage per container

Support & Training:

Included Support: • 6 months of technical support via email and WhatsApp • Quarterly system health checks • Emergency hotline for critical issues (24/7) • Regular updates and bug fixes **Training Provided:** • 2-day onsite training for finance team • Administrator training for IT staff • User manual and video tutorials • Monthly office hours for questions **Future Enhancements (Roadmap):** • Voice command interface • Mobile app (iOS/Android) • Advanced predictive analytics • Automated board presentation generation • Integration with accounting software (QuickBooks, Sage, etc.) • Multi-bank consolidation for holding companies

Technical Inquiries Welcome ■■■

Have technical questions? Want to discuss integration with your existing systems? Need custom features for your specific use case? I'm here to help! As the architect and developer of this system, I can provide detailed technical consultation, custom development, and ongoing support tailored to your needs.

Olaniyi Micheal Oluwatosin
AI Engineer | Python Developer | Financial Technology Specialist

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■ Location: Lagos, Nigeria

■ Available for:
• System customization and integration
• Technical consulting and architecture review
• AI/ML project development
• Remote AI engineering contracts
• International collaborations

Building the future of financial technology, one AI agent at a time. ■